



**DEMONSTRATE AN UNDERSTANDING OF
ISSUES AFFECTING THE MANAGEMENT
OF A LOCAL AREA COMPUTER
NETWORK (LAN)**

UNIT STANDARD 114046

NQF LEVEL: 5

CREDITS: 4

NOTIONAL HOURS: 40

LEARNER INFORMATION	
Name	
Surname	
ID Number	
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LEARNER GUIDE

Learner Information:

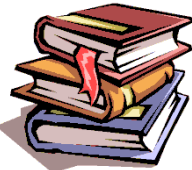
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Key to Icons

The following icons may be used in this Learner Guide to indicate specific functions:

 Books	This icon means that other books are available for further information on a particular topic/subject.
 References	This icon refers to any examples, handouts, checklists, etc...
 Important	This icon represents important information related to a specific topic or section of the guide.
 Activities	This icon helps you to be prepared for the learning to follow or assist you to demonstrate understanding of module content. Shows transference of knowledge and skill.
 Exercises	This icon represents any exercise to be completed on a specific topic at home by you or in a group.
 Tasks/Projects	An important aspect of the assessment process is proof of competence. This can be achieved by observation or a portfolio of evidence should be submitted in this regard.



**Workplace
Activities**

An important aspect of learning is through workplace experience. Activities with this icon can only be completed once a learner is in the workplace



Tips

This icon indicates practical tips you can adopt in the future.



Notes

This icon represents important notes you must remember as part of the learning process.

Learner Guide Introduction

About the Learner Guide...	This Learner Guide provides a comprehensive overview of the 114046 Demonstrate an understanding of issues affecting the management of a local area computer network (LAN) and forms part of a series of Learner Guides that have been developed. The series of Learner Guides are conceptualized in modular's format and developed FOR NATIONAL CERTIFICATE: INFORMATION TECHNOLOGY: SYSTEMS SUPPORT: SAQA 48573 -LEVEL 5- 147 CREDITS. They are designed to improve the skills and knowledge of learners, and thus enabling them to effectively and efficiently complete specific tasks. Learners are required to attend training workshops as a group or as specified by their organization. These workshops are presented in modules, and conducted by a qualified facilitator.
Purpose	114046 Demonstrate an understanding of issues affecting the management of a local area computer network (LAN)
Outcomes	• 114046 Demonstrate an understanding of issues affecting the management of a local area computer network (LAN)
Assessment Criteria	The only way to establish whether a learner is competent and has accomplished the specific outcomes is through an assessment process. Assessment involves collecting and interpreting evidence about the learner's ability to perform a task. This guide may include assessments in the form of activities, assignments, tasks or projects, as well as workplace practical tasks. Learners are required to perform tasks on the job to collect enough and appropriate evidence for their portfolio of evidence, proof signed by their supervisor that the tasks were performed successfully.
To qualify	To qualify and receive credits towards the learning programme, a registered assessor will conduct an evaluation and assessment of the learner's portfolio of evidence and competency
Range of Learning	This describes the situation and circumstance in which competence must be demonstrated and the parameters in which learners operate
Responsibility	The responsibility of learning rest with the learner, so: • Be proactive and ask questions, • Seek assistance and help from your facilitators, if required.

Demonstrate an understanding of issues affecting the management of a local area computer network (LAN)

Learning Unit **1**

UNIT STANDARD NUMBER	:	114046
LEVEL ON THE NQF	:	5
CREDITS	:	4
FIELD	:	Physical, Mathematical, Computer and Life Sciences
SUB FIELD	:	Information Technology and Computer Sciences

PURPOSE:

- This unit standard is intended:
- To provide conceptual knowledge of the areas covered;
- For those working in, or entering the workplace in the area of Data Communications & Networking.
- People credited with this unit standard are able to:
- Explain the management of access to a local area computer network (LAN)
- Explain LAN performance issues
- Explain LAN support issues
- Explain typical viruses on LANs
- The performance of all elements is to a standard that allows for further learning in this area.

LEARNING ASSUMED TO BE IN PLACE:

The credit value of this unit standard is calculated assuming a person has the prior knowledge and skills to:

Demonstrate an understanding of local area networks

Explain the principles of computer networks

Apply the principles of supporting users of a local area network

SPECIFIC OUTCOME 1.

Set up user access to a local area computer network.

Learning Outcomes

- 1. The explanation identifies resources whose access can be managed.
- 2. The explanation outlines the level of access for different categories of people.
- 3. The explanation outlines methods of controlling access.
- 4. The explanation outlines the purpose of an access audit trail.

1.1 Set up user access to a local area computer network.

PHYSICAL COMPONENTS

A local area network generally requires three principal components besides the computers being connected: network cards, cable or wire, and software. While software-driven, the physical properties of a LAN include interfaces, called network access units, which connect computers to networks. These units are actually network cards installed on computer motherboards. Their job is to provide a connection, monitor availability of access, set or buffer the data transmission speed, ensure against transmission errors and collisions, and assemble data from the LAN into usable form for the PC. A LAN consisting of two to four computers, however, can be created without a network card and this kind of network is called a slotless system. In a slotless system, the computers' serial and parallel ports are connected to each other. Such LANs are very inexpensive and businesses use them largely for sharing hard-drive space and printers, but they cannot support high-speed data transmission.

The next part of a LAN is the wiring, which provides the physical connection from one PC to another, and to servers and printers and other peripherals. The properties of the wiring determine transmission speeds.

The first LANs were connected with coaxial cable, a variety of the type used to deliver cable television. Certain kinds of coaxial cable are relatively inexpensive and coaxial cable is simple to attach. More importantly, these cables provide great bandwidth (the system's rate of data transfer), enabling transmission speeds up to 20 megabits per second.

During the 1980s, however, AT&T introduced a LAN wiring system using ordinary twisted wire pair of the type used for telephones. The primary advantages of twisted wire pair are that it is very cheap, simpler to splice than coaxial, and is already installed in many buildings as obsolete or redundant wiring. In fact, many buildings were left with stranded 25-pair wiring once used for key telephone systems.

But the downside of this simplicity is that its bandwidth is more limited, meaning that twisted pair, designed for voice communication, transmits data at a slow rate. For example, AT&T's first LAN product, StarLAN, had a capacity of only one megabit per second. Subsequent improvements expanded this capacity tenfold and eliminated the need for shielded, or conditioned, wiring.

A more recent development in LAN wiring is fiber distributed data interface (FDDI) or fiber-optic cable. This type of wiring uses thin strands of glass to transmit pulses of light between terminals. Its advantages are that it provides tremendous bandwidth and thus allows very high transmission speeds: data transmission at a rate of up to 100 megabits per second. And, because it is optical rather than electronic, it is impervious to electromagnetic interference. Fiber optics also supports a network of up to 1,000 computers and can transmit signals up to 50 miles. Its main drawbacks, however, are that splicing is difficult and requires a high degree of skill and that it costs far more than its counterparts.

The primary application of fiber is not between terminals, but between LAN buses (terminals) located on different floors. As a result, FDDI is used mainly in building risers. Within individual floors, LAN facilities remain coaxial or twisted wire pair.

Where a physical connection cannot be made, such as across a street or between buildings where easements for wiring cannot be secured, microwave radio may be used. It is often difficult, however, to secure frequencies for this medium.

Another alternative in this application is light transceivers, which project a beam of light similar to fiber-optic cable, but through the air, rather than over cable. These systems do not have the frequency allocation or radiation problems associated with microwave, but they are susceptible to interference from fog and other obstructions.

The software needed for a LAN depends on the kind of network being created: whether slotless, peer-to-peer, or server-based. Slotless system software usually enables users to perform the rudimentary tasks associated with slotless systems: sharing hard-drive space and printers. In addition, this software may provide e-mail and security features. Basic slotless system utilities are included in standard operating systems such as Windows 95 and Windows 98.

Peer-to-peer software facilitates peer-to-peer networks, which allow all users to access and use the resources of all computers attached to the network, including hard drives and printers. Peer-to-peer LANs, however, generally lack security and administration capabilities of server-based LANs, as well as the capacity for large data transmissions. Nevertheless, they are less expensive than their server-based counterparts.

Client/server software is designed for networks that designate a computer as the hub or server of the LAN. The server computer is linked to all the other computers of the network—the client computers—and it carries out the majority of the server duties, such as user access control and coordinating user tasks. The server cannot be used as a workstation, however, without special software allowing it to function as one. The client computers use the programs and data stored on the server. Server-based networks are best for companies with heavy network traffic.

TOPOLOGIES

LANs are designed in several different topologies or physical patterns of connecting terminals. The most common topology is the bus, where several terminals are connected directly to each other over a single transmission path. Its layout is linear and it resembles a street with several driveways. The bus network requires cables that allow signals to flow in either direction, called a full duplex medium. Each terminal on the bus LAN contends with other terminals for access to the system. When it has secured access to the system, it broadcasts its message to all the terminals at once. The message is picked up by the one terminal or group of terminal stations for which it is intended. The bus network's lack of routing and central control make it very reliable, because failure of one of the network's computers generally will not impede the flow of other network traffic.

A second topology, the star network, also works like a bus in terms of contention and broadcast. But in the star, stations are connected to a single, central node that administers access. The central node knows the path to all the other nodes, which makes routing easy. The central node also enables access control and establishing a priority status for users. Several of these nodes may be connected to one another. For example, a bus serving 6 stations may be connected to another bus serving 10 stations and a third bus connecting 12 stations. The star topology is most often used where the connecting facilities are coaxial or twisted wire pair.

The ring topology connects each station to its own node, and these nodes are connected in a circular fashion. Node 1 is connected to node 2, which is connected to node 3, and so on, and the final node is connected back to node 1. Messages sent over the LAN are regenerated by each node, but retained only by the addressees. Eventually, the message circulates back to the sending node, which removes it from the stream. Consequently, this configuration does not require routing.

1.2 The explanation outlines methods of controlling access.

Network access control (NAC), also called network admission control, is a method of bolstering the security of a proprietary network by restricting the availability of network resources to endpoint devices that comply with a defined security policy

Network access control (NAC), also called network admission control, is a method of bolstering the security of a proprietary network by restricting the availability of network resources to endpoint devices that comply with a defined security policy.

A traditional network access server (NAS) is a server that performs authentication and authorization functions for potential users by verifying logon information. In addition to these functions, NAC restricts the data that each particular user can access, as well as implementing anti-threat applications such as firewalls, antivirus software and spyware-detection programs. NAC also regulates and restricts the things individual subscribers can do once they are connected. Several major networking and IT vendors have introduced NAC products.

NAC is ideal for corporations and agencies where the user environment can be rigidly controlled. However, some administrators have expressed doubt about the practicality of NAC deployment in networks with large numbers of diverse users and devices, the nature of which constantly change. An example is a network for a large university with multiple departments, numerous access points and thousands of users with various backgrounds and objectives.

Controlling Network Access

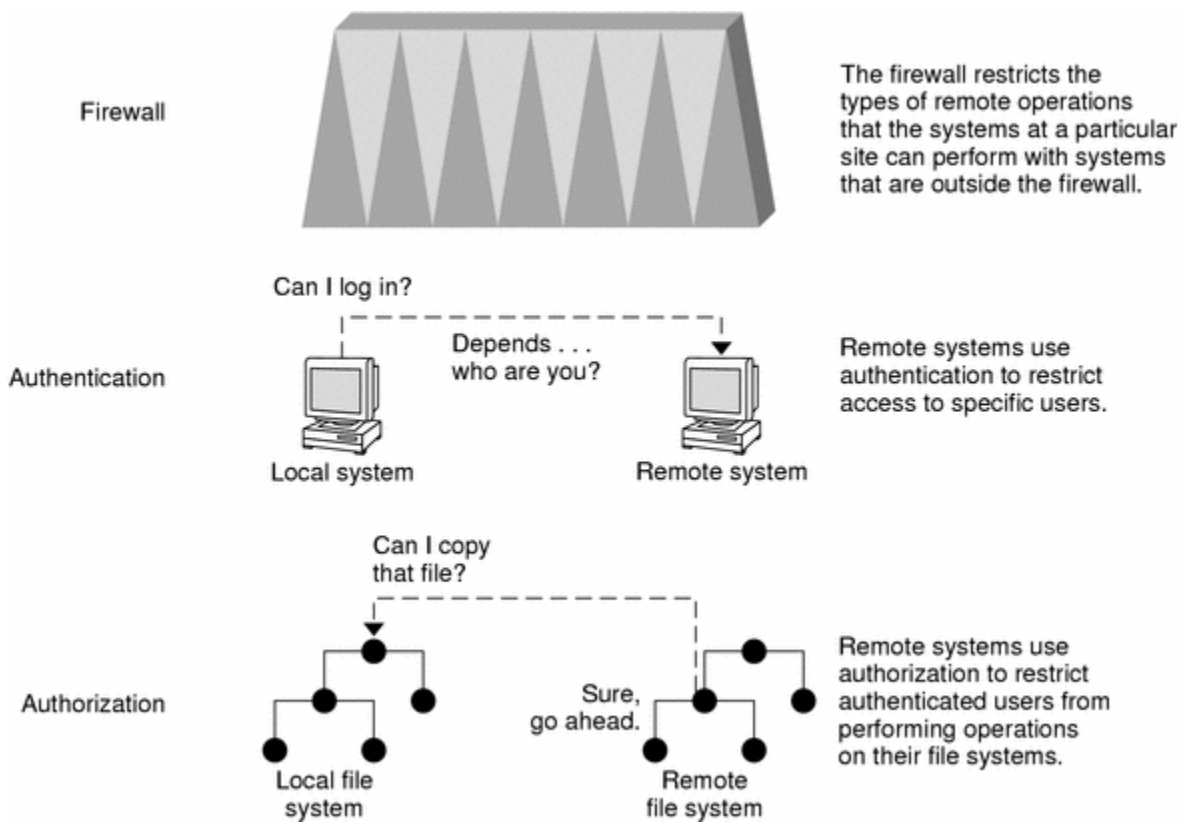
Computers are often part of a configuration of computers. The configuration is called a **network**. A network allows connected computers to exchange information. Networked computers can access data and other resources from other computers on the network. Networking has created a powerful and sophisticated way of computing. However, networking also jeopardizes computer security.

For instance, within a network of computers, individual machines are open to allow the sharing of information. Also, because many people have access to the network, unwanted access is more likely, especially through user error. For example, a poor use of passwords can allow unwanted access.

Network Security Mechanisms

Network security is usually based on limiting or blocking operations from remote systems. The following figure describes the security restrictions that you can impose on remote operations.

Figure 2–1 Security Restrictions for Remote Operations



1.3 The explanation outlines the purpose of an access audit trail.

AUDIT TRAILS

Audit trails maintain a record of system activity both by system and application processes and by user activity of systems and applications. In conjunction with appropriate tools and procedures, audit trails can assist in detecting security violations, performance problems, and flaws in applications. This bulletin focuses on audit trails as a technical control and discusses the benefits and objectives of audit trails, the types of audit trails, and some common implementation issues.

An audit trail is a series of records of computer events, about an operating system, an application, or user activities. A computer system may have several audit trails, each devoted to a particular type of activity. Auditing is a review and analysis of management, operational, and technical controls. The auditor can obtain valuable information about activity on a computer system from the audit trail. Audit trails improve the auditability of the computer system.

Audit trails may be used as either a support for regular system operations or a kind of insurance policy or as both of these. As insurance, audit trails are maintained but are not used unless needed, such as after a system outage. As a support for operations, audit trails are used to help system administrators ensure that the system or resources have not been harmed by hackers, insiders, or technical problems.

BENEFITS AND OBJECTIVES

Audit trails can provide a means to help accomplish several security-related objectives, including individual accountability, reconstruction of events (actions that happen on a computer system), intrusion detection, and problem analysis.

Individual Accountability

Audit trails are a technical mechanism that help managers maintain individual accountability. By advising users that they are personally accountable for their actions, which are tracked by an audit trail that logs user activities, managers can help promote proper user behavior. Users are less likely to attempt to circumvent security policy if they know that their actions will be recorded in an audit log.

For example, audit trails can be used in concert with access controls to identify and provide information about users suspected of improper modification of data (e.g., introducing errors into a database). An audit trail may record "before" and "after" versions of records. (Depending upon the size of the file and the capabilities of the audit logging tools, this may be very resource-intensive.) Comparisons can then be made between the actual changes made to records and what was expected. This can help management determine if errors were made by the user, by the system or application software, or by some other source.

Audit trails work in concert with logical access controls, which restrict use of system resources. Granting users access to particular resources usually means that they need that access to accomplish their job. Authorized access, of course, can be misused, which is where audit trail analysis is useful.

While users cannot be prevented from using resources to which they have legitimate access authorization, audit trail analysis is used to examine their actions. For example, consider a personnel office in which users have access to those personnel records for which they are responsible. Audit trails can reveal that an individual is printing far more records than the average user, which could indicate the selling of personal data. Another example may be an engineer who is using a computer for the design of a new product. Audit trail analysis could reveal that an outgoing modem was used extensively by the engineer the week before quitting. This could be used to investigate whether proprietary data files were sent to an unauthorized party.

Reconstruction of Events

Audit trails can also be used to reconstruct events after a problem has occurred. Damage can be more easily assessed by reviewing audit trails of system activity to pinpoint how, when, and why normal operations ceased. Audit trail analysis can often distinguish between operator-induced errors (during which the system may have performed exactly as instructed) or system-created errors (e.g., arising from a poorly tested piece of replacement code). If, for example, a system fails or the integrity of a file (either program or data) is questioned, an analysis of the audit trail can reconstruct the series of steps taken by the system, the users, and the application. Knowledge of the conditions that existed at the time of, for example, a system crash, can be useful in avoiding future outages. Additionally, if a technical problem occurs (e.g., the corruption of a data file) audit trails can aid in the recovery process (e.g., by using the record of changes made to reconstruct the file).

Intrusion Detection

Intrusion detection refers to the process of identifying attempts to penetrate a system and gain unauthorized access. If audit trails have been designed and implemented to record appropriate information, they can assist in intrusion detection. Although normally thought of as a real-time effort, intrusions can be detected in real time, by examining audit records as they are created (or through the use of other kinds of warning flags/notices), or after the fact (e.g., by examining audit records in a batch process).

Real-time intrusion detection is primarily aimed at outsiders attempting to gain unauthorized access to the system. It may also be used to detect changes in the system's performance indicative of, for example, a virus or worm attack (forms of malicious code). There may be difficulties in implementing real-time auditing, including unacceptable system performance.

After-the-fact identification may indicate that unauthorized access was attempted (or was successful). Attention can then be given to damage assessment or reviewing controls that were attacked.

Problem Analysis

Audit trails may also be used as on-line tools to help identify problems other than intrusions as they occur. This is often referred to as real-time auditing or monitoring. If a system or application is deemed to be critical to an organization's business or mission, real-time auditing may be implemented to monitor the status of these processes (although, as noted above, there can be difficulties with real-time analysis). An analysis of the audit trails may be able to verify that the system operated normally (i.e., that an error may have resulted from operator error, as opposed to a system-originated error). Such use of audit trails may be complemented by system performance logs. For example, a significant increase in the use of system resources (e.g., disk file space or outgoing modem use) could indicate a security problem.

AUDIT TRAILS AND LOGS

A system can maintain several different audit trails concurrently. There are typically two kinds of audit records, (1) an event-oriented log and (2) a record of every keystroke, often called keystroke monitoring. Event-based logs usually contain records describing system events, application events, or user events. An audit trail should include sufficient information to establish what events occurred and who (or what) caused them. In general, an event record should specify when the event occurred, the user ID associated with the event, the program or command used to initiate the event, and the result. Date and time can help determine if the user was a masquerader or the actual person specified.

Keystroke Monitoring

Keystroke monitoring is the process used to view or record both the keystrokes entered by a computer user and the computer's response during an interactive session. Keystroke monitoring is usually considered a special case of audit trails. Examples of keystroke monitoring would include viewing characters as they are typed by users, reading users' electronic mail, and viewing other recorded information typed by users.

Some forms of routine system maintenance may record user keystrokes. This could constitute keystroke monitoring if the keystrokes are preserved along with the user identification so that an administrator could determine the keystrokes entered by specific users. Keystroke monitoring is conducted in an effort to protect systems and data from intruders who access the systems without authority or in excess of their assigned authority. Monitoring keystrokes typed by intruders can help administrators assess and repair damage caused by intruders.

Audit Events

System audit records are generally used to monitor and fine-tune system performance. Application audit trails may be used to discern flaws in applications, or violations of security policy committed within an application. User audits records are generally used to hold individuals accountable for their actions. An analysis of user audit records may expose a variety of security violations, which might range from simple browsing to attempts to plant Trojan horses or gain unauthorized privileges.

The system itself enforces certain aspects of policy (particularly system-specific policy) such as access to files and access to the system itself. Monitoring the alteration of systems configuration files that implement the policy is important. If special accesses (e.g., security administrator access) have to be used to alter configuration files, the system should generate audit records whenever these accesses are used.

Sometimes a finer level of detail than system audit trails is required. Application audit trails can provide this greater level of recorded detail. If an application is critical, it can be desirable to record not only who invoked the application, but certain details specific to each use. For example, consider an e-mail application.

It may be desirable to record who sent mail, as well as to whom they sent mail and the length of messages. Another example would be that of a database application. It may be useful to record who accessed what database as well as the individual rows or columns of a table that were read (or changed or deleted), instead of just recording the execution of the database program.

A user audit trail monitors and logs user activity in a system or application by recording events initiated by the user (e.g., access of a file, record or field, use of a modem). Flexibility is a critical feature of audit trails. Ideally (from a security point of view), a system administrator would have the ability to monitor all system and user activity, but could choose to log only certain functions at the system level, and within certain applications. The decision of how much to log and how much to review should be a function of application/data sensitivity and should be decided by each functional manager/application owner with guidance from the system administrator and the computer security manager/officer, weighing the costs and benefits of the logging. Audit logging can have privacy implications; users should be aware of applicable privacy laws, regulations, and policies that may apply in such situations.

System-Level Audit Trails

If a system-level audit capability exists, the audit trail should capture, at a minimum, any attempt to log on (successful or unsuccessful), the log-on ID, date and time of each log-on attempt, date and time of each log-off, the devices used, and the function(s) performed once logged on (e.g., the applications that the user tried, successfully or unsuccessfully, to invoke). System-level logging also typically includes information that is not specifically security-related, such as system operations, cost-accounting charges, and network performance.

Application-Level Audit Trails

System-level audit trails may not be able to track and log events within applications, or may not be able to provide the level of detail needed by application or data owners, the system administrator, or the computer security manager. In general, application-level audit trails monitor and log user activities, including data files opened and closed, specific actions, such as reading, editing, and deleting records or fields, and printing reports.

Some applications may be sensitive enough from a data availability, confidentiality, and/or integrity perspective that a "before" and "after" picture of each modified record (or the data element(s) changed within a record) should be captured by the audit trail.

User Audit Trails

User audit trails can usually log:

- all commands directly initiated by the user;
- all identification and authentication attempts; and
- files and resources accessed.

It is most useful if options and parameters are also recorded from commands. It is much more useful to know that a user tried to delete a log file (e.g., to hide unauthorized actions) than to know the user merely issued the delete command, possibly for a personal data file.

IMPLEMENTATION ISSUES

Audit trail data requires protection, since the data should be available for use when needed and is not useful if it is not accurate. Also, the best planned and implemented audit trail is of limited value without timely review of the logged data. Audit trails may be reviewed periodically, as needed (often triggered by occurrence of a security event), automatically in real-time, or in some combination of these. System managers and administrators, with guidance from computer security personnel, should determine how long audit trail data will be maintained -- either on the system or in archive files.

Following are examples of implementation issues that may have to be addressed when using audit trails.

Protecting Audit Trail Data

Access to on-line audit logs should be strictly controlled. Computer security managers and system administrators or managers should have access for review purposes; however, security and/or administration personnel who maintain logical access functions may have no need for access to audit logs.

It is particularly important to ensure the integrity of audit trail data against modification. One way to do this is to use digital signatures. Another way is to use write-once devices. The audit trail files need to be protected since, for example, intruders may try to "cover their tracks" by modifying audit trail records. Audit trail records should be protected by strong access controls to help prevent unauthorized access. The integrity of audit trail information may be particularly important when legal issues arise, such as when audit trails are used as legal evidence. (This may, for example, require daily printing and signing of the logs.) Questions of such legal issues should be directed to the cognizant legal counsel.

The confidentiality of audit trail information may also be protected, for example, if the audit trail is recording information about users that may be disclosure-sensitive such as transaction data containing personal information (e.g., "before" and "after" records of modification to income tax data). Strong access controls and encryption can be particularly effective in preserving confidentiality.

Review of Audit Trails

Audit trails can be used to review what occurred after an event, for periodic reviews, and for real-time analysis. Reviewers should know what to look for to be effective in spotting unusual activity. They need to understand what normal activity looks like. Audit trail review can be easier if the audit trail function can be queried by user ID, terminal ID, application name, date and time, or some other set of parameters to run reports of selected information.

Audit Trail Review After an Event. Following a known system or application software problem, a known violation of existing requirements by a user, or some unexplained system or user problem, the appropriate system-level or application-level administrator should review the audit trails. Review by the application/data owner would normally involve a separate report, based upon audit trail data, to determine if their resources are being misused.

Periodic Review of Audit Trail Data. Application owners, data owners, system administrators, data processing function managers, and computer security managers should determine how much review of audit trail records is necessary, based on the importance of identifying unauthorized activities. This determination should have a direct correlation to the frequency of periodic reviews of audit trail data.

Real-Time Audit Analysis. Traditionally, audit trails are analyzed in a batch mode at regular intervals (e.g., daily). Audit records are archived during that interval for later analysis. Audit analysis tools can also be used in a real-time, or near real-time fashion. Such intrusion detection tools are based on audit reduction, attack signature, and variance techniques. Manual review of audit records in real-time is almost never feasible on large multiuser systems due to the volume of records generated. However, it might be possible to view all records associated with a particular user or application, and view them in real time. (This is similar to keystroke monitoring, though, and may be legally restricted.)

Tools for Audit Trail Analysis

Many types of tools have been developed to help to reduce the amount of information contained in audit records, as well as to distill useful information from the raw data. Especially on larger systems, audit trail software can create very large files, which can be extremely difficult to analyze manually. The use of automated tools is likely to be the difference between unused audit trail data and a robust program. Some of the types of tools include:

Audit reduction tools are preprocessors designed to reduce the volume of audit records to facilitate manual review. Before a security review, these tools can remove many audit records known to have little security significance. (This alone may cut in half the number of records in the audit trail.) These tools generally remove records generated by specified classes of events, such as records generated by nightly backups might be removed.

Trends/variance-detection tools look for anomalies in user or system behavior. It is possible to construct more sophisticated processors that monitor usage trends and detect major variations. For example, if a user typically logs in at 9 a.m., but appears at 4:30 a.m. one morning, this may indicate a security problem that may need to be investigated. Attack signature-detection tools look for an attack signature, which is a specific sequence of events indicative of an unauthorized access attempt. A simple example would be repeated failed log-in attempts.

COST CONSIDERATIONS

Audit trails involve many costs. First, some system overhead is incurred recording the audit trail. Additional system overhead will be incurred storing and processing the records. The more detailed the records, the more overhead is required. Another cost involves human and machine time required to do the analysis. This can be minimized by using tools to perform most of the analysis. Many simple analyzers can be constructed quickly (and cheaply) from system utilities, but they are limited to audit reduction and identifying particularly sensitive events. More complex tools that identify trends or sequences of events are slowly becoming available as off-the-shelf software. (If complex tools are not available for a system, development may be prohibitively expensive. Some intrusion detection systems, for example, have taken years to develop.) The final cost of audit trails is the cost of investigating anomalous events. If the system is identifying too many events as suspicious, administrators may spend undue time reconstructing events and questioning personnel.

In groups outline the level of access for different categories of people.

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SPECIFIC OUTCOME 2.

Explain local area computer network performance issues.

Learning Outcomes

- 1. The explanation describes a range of factors that affects response times on a LAN.
- 2. The explanation outlines the need to analyse data and identify problems.
- 3. The explanation outlines how diagnostic tools are used to collect data.
- 4. The explanation outlines and compares methods for improving the performance with respect to their effect on performance.

2.1 Explain local area computer network performance issues.

Local Area Networks (LANs) and Wide Area Networks (WANs) have much in common, but the differences are enough to make them two separate acronyms in speech and in practice. Each one has its benefits and downsides, and these advantages and disadvantages can affect an organization's productivity significantly. So what exactly is the difference between a LAN connection and a WAN connection?

Local Area Networks (LANs)

Advantages	Disadvantages
• Speed • Cost • Ease of Setup	• Limited to Small Area

LAN connections can only operate in a local area which is usually not any bigger than a house, or a floor in an office building. Typically a LAN will consist of only a handful of clients, but can have upwards of a hundred.

One of the major advantages with LANs are the speeds they can reach. With a LAN, it isn't uncommon to see technology ready for 1Gbps (1 gigabit per second). **Example:** If you were to download all 3 816 000 English articles off of Wikipedia it would take just over 13 hours to do so, where as a WAN would take 16 days to do the same. A LAN can operate up to 30x faster than a WAN

Another advantage to having a LAN connection is the cost. It is relatively cheap to have as it tends to require less hassle to set up and less advanced infrastructure to keep it running. This is mainly due to the technological components of a LAN, which brings me to the next difference.

A LAN connection tends to require some relatively simple things to set it up. All you need is some Ethernet cables, a network switch, and you are good to go. Alternatively, you can also see LANs being done over Wi-Fi, or you can use Wi-Fi in conjunction with standard Ethernet connections to create a LAN available to all kinds of devices, whether it be a smartphone or a desktop computer.

The major disadvantage with a LAN is inherent in its name. "Local" Area Networks are only good as far as you can reach an Ethernet cable or Wi-Fi signal. Simply put, you cannot buy an Ethernet cable that will reach throughout an entire building, and a Wi-Fi connection rapidly deteriorates as you get further than a few dozen meters away.

DIFFICULTIES

LANs are susceptible to many kinds of transmission errors. Electromagnetic interference from motors, power lines, and sources of static, as well as shorts from corrosion, can corrupt data. In addition, different kinds of cables are more susceptible to these problems than others. Software bugs and hardware failures can also introduce errors, as can irregularities in wiring and connections.

LANs generally compensate for these errors by working off an uninterruptible power source, such as batteries, and using backup software to recall most recent activity and hold unsaved material. Some systems may be designed for redundancy, such as keeping two file servers and alternate wiring to route around failures.

In addition, as computer software evolves requiring faster processors and faster rates of transmission, LAN technology also must evolve. Multimedia and video applications in particular force companies to upgrade their LANs in order to use such applications in a network environment. Consequently, LANs increasingly need to transmit data at gigabit, not megabit, speeds and hence older technology must be upgraded or replaced.

PURCHASING A LAN

When purchasing a LAN, or even investigating the possibility of installing one, several considerations must be kept in mind. The **costs** involved and the administrative support needed often far exceed reasonable predictions.

Three general concerns when considering a LAN include administration, security, and productivity. Administration utilities regulate and coordinate file, application, peripheral, and resource use, while security utilities control access to the network. Productivity refers to the tasks a company wants to perform via a LAN, which may include file, database, and printer sharing.

Moreover, thorough consideration of potential costs should include such factors as purchase price of equipment, spare parts and **taxes**, installation costs, labor and building modifications, and permits. Operating costs include forecasted public network traffic, diagnostics, and routine maintenance. In addition, the buyer should seek a schedule of potential costs associated with upgrades and expansion of the network, since company LANs tend to require new technology and to expand periodically.

The vendor should agree to a **contract** expressly detailing the degree of support that will be provided in installing and turning on the system. In addition, the vendor should provide a maintenance contract that binds the company to make immediate, free repairs when performance of the system exceeds prescribed standards. All of these factors should be addressed in the buyer's **request for proposal**, which is distributed to potential vendors.

SPECIFIC OUTCOME 3.

Explain local area computer network support issues.

Learning Outcomes

- 1. The explanation distinguishes sources.
- 2. The explanation outlines user expectations of a range of support options.

3.1 Explain local area computer network support issues

TRANSMISSION METHODS USED BY LANs

LANs are effective because their transmission capacity is greater than any single terminal on the system. As a result, each station terminal can be offered a certain amount of time on the LAN, like a time-sharing arrangement. To take advantage of this window of opportunity, stations organize their messages into compact packets that can be quickly disseminated. In contending for access, a station with something to send stores its data packet in a buffer until the LAN is clear. At that point the message is sent out. Sometimes, two stations may detect the opening at the same time and send their messages simultaneously. Unaware that another message has been sent out, the two signals will collide on the LAN. When this happens it is up to the software to determine who should go first and ask both machines to try again. In busy LANs, collisions would occur all the time, slowing the system down considerably. To solve the problem, the LAN software circulates a token. This works like a ticket that is distributed only to one station at a time. Instead of waiting for the LAN to clear, the station waits to receive the token. When it has the token, the station sends its packet out over the LAN. When it is done, it returns the token to the stream for the next user. Tokens, used in ring and bus topologies, virtually eliminate the problem of collisions by providing orderly, noncontention access.

The transmission methods used on LANs are either baseband or broadband. The baseband medium uses a high-speed digital signal consisting of square wave DC voltage. While it is fast, it can accommodate only one message at a time. As a result it is suitable for smaller networks where contention is low. It also is very simple, requiring no tuning or frequency discretion circuits. As a result, the transmission medium may be connected directly to the network access unit and is suitable for use over twisted wire pair facilities.

In contrast, the broadband medium tunes signals to special frequencies, much like cable television. Stations are instructed by signalling information to tune to a specific channel to receive information. The information within each channel on a broadband medium may also be digital, but they are separated from other messages by frequency. As a result, the medium generally requires higher capacity cables, such as coaxial cable. Suited for busier LANs, broadband systems require the use of tuning devices in the network access unit that can filter out all but the single channel it needs.

SERVERS

File and printer servers provided the initial impetus for companies to develop LANs so that they could share databases and expensive peripherals such as printers. Furthermore, the heart of the LAN, the administrative software, generally resides either in a dedicated file server (which functions as a server only) or, in a smaller, less busy LAN, in a computer acting as a file server (which also can function as a workstation). In addition to acting as a kind of traffic cop by controlling and regulating user access, this server holds files for shared use in its hard drives, administers applications such as operating systems, and coordinates tasks such as printing. Where a single computer is used both as a workstation and a file server, response times may lag because its processors are forced to perform several instructions at once. In addition, the system will store certain files on different computers connected with the LAN. Consequently, if one machine is down, the entire system may be crippled. Moreover, if the system were to crash due to under capacity, some data may be lost or corrupted.

The addition of a dedicated file server may be costly, but it provides several advantages over a distributed system. In addition to ensuring access even when some machines are down, it is unencumbered by multiple duties. Its only jobs are to hold files and provide access. Since 1990 one of the most notable developments in LANs has been the growth of communication servers that allow LANs to communicate with networks outside of the LANs themselves. Communication servers enable remote LAN access, e-mail, fax, and other communication services. Like other servers, this one controls access and facilitates use of communications software and hardware. As with file and printer servers, a separate computer may be designated as a dedicated server to enhance reliability. Furthermore, the LANs of the late

1990s began to include servers devoted other applications such as those for decision support, transaction processing, and data warehousing. The number of application servers is forecast to increase significantly as more companies add dedicated application servers to their LANs.

SPEED MEASUREMENTS

The speed of the LAN is measured in terms of throughput, a figure different from transmission speed because it takes into account the capacity of the wiring and the distance between stations. The data rate, which most directly represents response time, is determined by throughput and other factors such as overhead bits and other signals, error and collision recovery, software and hardware efficiency, and the memory capacity of disk drives.

OTHER EQUIPMENT

As mentioned earlier, LANs are generally limited in size because of the physical properties of the network: distance, impedance (a kind of electrical resistance), and load. Some equipment, such as repeaters, can extend the range of a LAN. Repeater have no processing ability, but simply regenerate signals that are weakened by impedance. Other types of LAN equipment with processing ability include gateways, which refer to the hardware and software necessary to enable technologically different networks to communicate with each other. A gateway, for example, can compensate for dissimilar protocols to pass information by translating them into a simpler code, such as ASCII. A bridge works like a gateway, but instead of connecting technologically different networks, it connects networks employing the same kind of technology. Similar to a bridge, a router is the hardware and software connection between two (or more) networks or subnetworks that routes traffic from one network or subnetwork to another. But routers primarily control the transmission of packets to their destinations.

Gateways, bridges, and routers can act as repeaters, boosting signals over greater distances. They also enable separate LANs located in different buildings to communicate with each other.

In some cases, separate LANs located in different cities—and even separate countries—may be linked over a public network. Whether these are "nailed up" dedicated links or switched services, the connection of two or more such LANs in separate geographic locations is referred

to as a wide area network (WAN). WANs require the use of special software programs in the operating system to enable dial-up connections that may be performed by a router. Unless limited to modem speeds, these connections may require special services, such as integrated services digital network (ISDN), to ensure efficient transmission, particularly of large data files. Increasingly, companies employing LANs in separate locations also operate WANs.

Another device, which can be used to create LANs, is the private branch exchange. Private branch exchanges (PBXs) are telephone switching systems that generally serve one company or network and route data and information to specific servers, rather than broadcasting to all stations. PBXs are oblivious to operating systems and use only twisted pair. As a result, PBX networks are somewhat slower and their applications are more limited than other kinds of LANs.

SPECIFIC OUTCOME 4.

Explain typical viruses on local area computer networks.

Learning Outcomes

- 1. The explanation outlines the symptoms and transmission of viruses.
- 2. The explanation allows the selection of a method for the prevention, detection, and eradication of viruses for a situation.

4.1 Explain typical viruses on local area computer networks.

Types of Computer Viruses

What Is a Computer Virus

There are all **types of computer viruses** but what is a computer virus? A computer virus has been defined as a set of computer instructions that reproduces itself and it may attach to other executable code. Usually this code is a short program that may either embed in other code or stand on it's own. In essence, this computer program is designed to infect some aspect of the host computer and then copy itself as much and as often as it has the chance. It was estimated that a virus by the name of mydoom infected well over a quarter a million computers in one day back in 2004. There are tens of thousands of worms and viruses now being spread via the internet with new ones being discovered each and every day. It is often through quite innocuous and normal internet activities like the exchange of files like music, photos and others that many people are infected with these unwanted and sometimes dangerous programs.

The nomenclature that is now used to describe viruses has changed considerably over the last few years as more, if not most, computers are now on the internet. This has ushered in a change in the **types of computer viruses** toward a worm/virus hybrid and has caused the distinction between them to vanish. There are a whole group of people that spend a tremendous amount of time looking for what have been termed backdoors into your computer so they can find ways to inject their code into your computer and use it for their own intentions.

What Does A Virus Do

The task of a virus is not always destructive like deleting files that may be important or something like causing your hard drive to crash. Many viruses these days are more interested in harvesting information from your computer and or using it as a zombie for their intentions like spam or other illegal purposes. In times past it was often the intention of a virus to do damage just for the sake of destruction and maybe bragging rights among peers but almost all viruses today that are widespread have at their root some economic agenda.

How Does A Virus Spread

One of the intents of all **types of computer viruses** that gets installed on your computer will be to spread itself. This happens in a fashion that is not all that different from what happens with a virus in the human population. It is through exposure that the virus spreads when the computers defenses are down or non existent. Also like their biological counterpart the computer virus can be spread rapidly and are not very easy to get rid of. Because the way a virus operates is to be stealthy, coupled with the rapid communications that happen between today's computers, it is easy to dramatically increase the speed at which a virus that is targeting a newly discovered vulnerability can move around the web and around the world.

A virus that targets a network of computers can even more easily spread since so many computers are connected and most likely will have the same vulnerability and easy access to one another. Often viruses will spread via shared folders, email or over other media that is often exposed to other computers via removable media like cds and flash drives.

Because there are so many types of computer viruses, a virus can infect another computer unintentionally anytime that program is run and the virus is activated. Something like opening a email attachment or downloading a file off the internet or giving or receiving a copy of a program or file from a co-workers thumb drive can expose you and others to a computer virus. Literally the gamut of these types of computer viruses can expose you whenever you have a downloaded file or a external drive attached to your pc. The most common way they spread is via email attachments or with the use or transfer of files via instant messaging.

The Types of Computer Viruses

There are six broad categories or types of computer viruses:

1. **Boot Sector Virus**
2. **File Infection Virus**
3. **Multipartite Virus**
4. **Network Virus**
5. **E-mail Virus**
6. **Macro Virus**

File viruses: File-infecting viruses attack executable programs, such as all files with “.exe” and “.com” extensions.

Script viruses: Script viruses are a subset of file viruses, written in a variety of script languages (VBS, JavaScript, BAT, PHP, etc.). They either infect other scripts (for example, Windows or Linux command and service files), or form a part of multi-component viruses. Script viruses are able to infect other file formats, such as HTML, if the file format allows the execution of scripts.

Boot viruses: Boot viruses attack boot sectors (removable media boot sector or hard disk master boot sector) and set their own loading routines at start-up

Macro viruses: Macro viruses attack documents where other commands (macros) can be inserted. These viruses are often embedded within word processing or spreadsheet applications, since macros are easily inserted into these types of files.

Viruses can also be classified according to the way they perform their action. While **direct action viruses** perform an action immediately after the infected object is activated, **resident viruses** stay and work in the computer's memory.

Malware

Malware is the term used to describe malicious programs and techniques that are not viruses but still pose a threat to your system.

Worm: A worm is an independent program that copies itself across a network. Unlike a virus (which needs the infected file to be copied in order to replicate itself), the worm spreads actively by sending copies of itself via LAN or Internet, email communication, or through operating system security bugs.

They can also bring with them additional malware (such as installing backdoor programs—see below), though this behavior is not strictly limited to worms. Worms can cause a great deal of damage—often they are used to “jam” communication channels by means of a DoS attack. A worm is capable of spreading worldwide, via the Internet, in minutes.

Trojan: A Trojan is a malware program that, unlike viruses or worms, cannot copy itself and infect files. It is usually found in the form of an executable file (.exe, .com) and does not contain anything aside from the Trojan code itself. For this reason, the only solution is to delete it. Trojans have various functions, from keylogging (they log and transmit every keystroke), to deleting files or disc formatting. Some contain a special feature that installs a backdoor program, (a client-server application that grants the developer remote access to your computer). Unlike common (legitimate) software with similar functions, it installs itself without the consent of the client computer.

Adware: Adware is short for advertising-supported software, which is software dedicated to displaying advertisements. Adware works by displaying pop-up windows during Internet browsing, by setting various websites as your homepage or by opening a special program interface window. Adware is often bundled with free-to-download programs, and the client is usually informed of this in the End User License Agreement. Adware advertisements allow freeware developers to earn revenue by offering program features available only with the paid version. In most cases, installation of adware falls within legal guidelines—there are many legitimate advertising-supported programs. However, issues such as the assertiveness of advertisements as well as their content can make the legality of some adware questionable.

Spyware: Spyware is software that uses the Internet for collecting various pieces of sensitive information about the user without his/her awareness. Some Spyware programs search for information such as currently installed applications and a history of visited websites. Other Spyware programs are created with a far more dangerous aim: the collection of financial or personal data for the purpose of Identity Theft.

Riskware: This type of malware includes all applications that increase the user's security risk when running. As with spyware and adware installation, riskware installation may be confirmed by license agreement. "Dialers" are a common example of Riskware—programs that divert connection to a preset paid number. Such programs can be legally used for Internet service payments, but they are often misused and the diverting occurs without the user's awareness.

Dangerous applications: A dangerous application is the term used for legal programs that, though installed by the user, may subject him/her to security risks. Examples include commercial keylogging or screen capture, remote access tools, password-cracking and security testing programs.

Hoaxes: A hoax is deliberate misinformation sent by email and spread with the help of an unsuspecting or uninformed public. Hoaxes are typically designed to get a user to do something they should not do. Malicious hoaxes often advise users to delete valid operating system files, claiming that the file is a dangerous virus.

In many cases, hoaxes refer to a credible institution/company in order to gain the reader's attention. For example, "Microsoft warns that..." or "CNN announced". These messages often warn of disastrous or even catastrophic consequences. The warnings have one thing in common – they urge users to send the messages to everyone they know, which perpetuates the life-cycle of the hoax. 99.9% of these types of messages are hoaxes.

Hoaxes cannot spread by themselves, the only way to protect yourself is to verify the authenticity of an email message's claims before taking any action.

Scams: Broadly defined, scams are deceptions perpetrated on computer users for the purpose of financial gain or identify theft. One of the most common scams involves an unsolicited fax, email, or letter from Nigeria or other West-African nation. The letter will appear to be a legitimate business proposal, but will require an advanced fee from the target. The proposal is of course fraudulent, and any fees paid by the target are immediately stolen.

Another common form of scamming includes phishing email messages and websites. The purpose of these scams is to gain access to sensitive data such as bank account numbers, PIN codes, etc. Access is usually achieved by sending email masquerading as a trustworthy person or business (financial institution, insurance company).

The email (or website that the user is directed to) can look very genuine and will contain graphics and content that may have originally come from the source that it is impersonating. The user will be asked to enter personal data such as bank account numbers or usernames and passwords. All such data, if submitted, can easily be stolen and misused.

Remote attacks

Special techniques which allow attackers to compromise remote systems. These are divided into several categories:

DoS attacks: DoS, or Denial of Service, is an attempt to make a computer or network unavailable for its intended users. DoS attacks obstruct communications between affected users, preventing them from continuing in a functional way. One common method of attack involves saturating the target machine with external communications requests, so that the target machine cannot respond to legitimate traffic, or responds so slowly as to be rendered effectively unavailable. Such attacks usually lead to a server overload. Computers exposed to DoS attacks usually need to be restarted in order to work properly. The targets of DoS attacks are web servers and the aim is to make them unavailable to users for a certain period of time.

DNS Poisoning: Using DNS (Domain Name Server) poisoning, hackers can trick the DNS server of any computer into believing that fake data is legitimate and authentic. The fake information is cached for a certain period of time, allowing attackers to rewrite DNS replies of IP addresses. As a result, users trying to access DNS poisoned websites will download computer viruses or worms instead of the website's original content.

Port scanning: Port scanning is used to determine which computer ports are open on a network host. A port scanner is software designed to find such ports. A computer port is a virtual point which handles incoming and outgoing data – this is crucial from a security point of

view. In a large network, the information gathered by port scanners may help to identify potential vulnerabilities. Such use is legitimate. Still, port scanning is often used by hackers attempting to compromise security. Their first step is to send packets to each port. Depending on the response type, it is possible to determine which ports are in use. The scanning itself causes no damage, but be aware that this activity can reveal potential vulnerabilities and allow attackers to take control of remote computers. Network administrators are advised to block all unused ports and protect those that are in use from unauthorized access.

TCP desynchronization: TCP desynchronization is a technique used in TCP Hijacking attacks. It is triggered by a process in which the sequential number in incoming packets differs from the expected sequential number. Packets with an unexpected sequential number are dismissed (or saved in buffer storage if they are present in the current communication window). In desynchronization, both communication endpoints dismiss received packets, at which point remote attackers are able to infiltrate and supply packets with a correct sequential number. The attackers can even manipulate or modify communication. TCP Hijacking attacks aim to interrupt server-client and/or peer-to-peer communications. Many attacks can be avoided by using authentication for each TCP segment. It is also advised to use the recommended configurations for your network devices.

SMB Relay: SMBRelay and SMBRelay2 are special programs that are capable of carrying out attacks against remote computers. The programs take advantage of the Server Message Block file sharing protocol which is layered into NetBIOS. A user sharing any folder or directory within the LAN most likely uses this file sharing protocol. Within local network communication, password hashes are exchanged.

SMBRelay receives a connection on UDP port 139 and 445, relays the packets exchanged by the client and server, and modifies them. After connecting and authenticating, the client is disconnected. SMBRelay creates a new virtual IP address. The new address can be accessed using the command "net use \\192.168.1.1". The address can then be used by any of the Windows networking functions. SMBRelay relays SMB protocol communication except for negotiation and authentication. Remote attackers can use the IP address as long as the client computer is connected.

SMBRelay2 works on the same principle as SMBRelay, except it uses NetBIOS names rather than IP addresses. Both can carry out "man-in-the-middle" attacks. These attacks allow remote attackers to read, insert and modify messages exchanged between two communication endpoints without being noticed. Computers exposed to such attacks often stop responding or restart unexpectedly. To avoid attacks we recommend that you use authentication passwords or keys.

ICMP attacks: ICMP (Internet Control Message Protocol) is a popular and widely-used Internet protocol. It is used primarily by networked computers to send various error messages. Remote attackers attempt to exploit the weaknesses of ICMP protocol. ICMP protocol is designed for one-way communication requiring no authentication. This enables remote attackers to trigger DoS (Denial of Service) attacks, or attacks which give unauthorized individuals access to incoming and outgoing packets. Typical examples of an ICMP attack are ping flood, ICMP_ECHO flood and smurf attacks. Computers exposed to an ICMP attack will experience significantly slower performance in applications that use the Internet and have problems connecting to the Internet.

Probes

Probes or *port scanners* check for improperly secured servers or services that may be running on computers on your LAN (especially the one that is directly connected to the Internet). These checks are usually performed by programs that take a range of IP addresses selected by the person running the program, and look for common services like Web, mail, FTP, Telnet, pcAnywhere, or proxy servers. If any of these (or other) services are found, the program tries to see if it can login or otherwise gain access to that service. If it can, it flags that IP address and service to the person running the program and what happens after that depends on what they have in mind for your system! Your best defence against probes is to not run any servers or services that you don't understand, or are not sure what they are used for. You also should properly secure any services that you do run. Finally, if you are a relative "newbie" and/or especially paranoid about intruders, *don't run the free Wingate 2.1d version*, which is easily misconfigured and can allow intruders into your LAN.